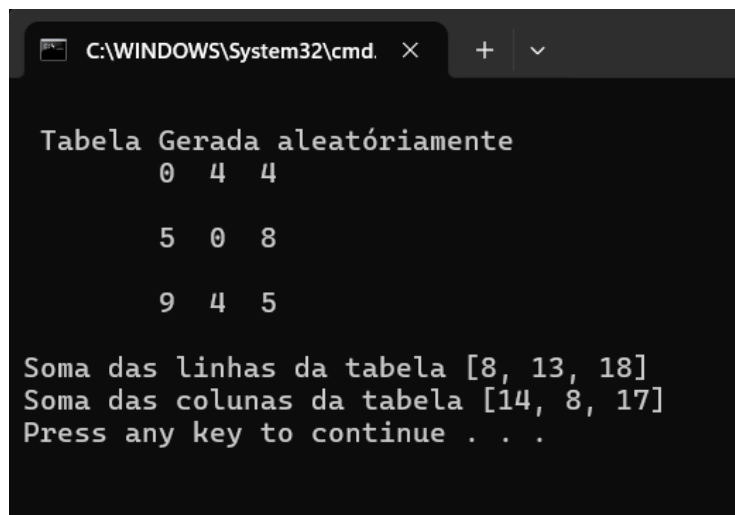


Algorithmics and Data Structures

UNIDADE CURRICULAR

Sheet 05 – Multidimensional lists

1. Implement a program that allows you to generate a two-dimensional 3x3 list, that is, a matrix with 3 rows and 3 columns.
 - Fill the list with random integer values between 0 and 9;
 - Call a **showTable()** function that displays the generated table;
 - Call a **verifyTable()** function that checks and prints the sum of each row and column.



```
C:\WINDOWS\System32\cmd. X + v

Tabela Gerada aleatoriamente
  0  4  4
  5  0  8
  9  4  5

Soma das linhas da tabela [8, 13, 18]
Soma das colunas da tabela [14, 8, 17]
Press any key to continue . . .
```

2. Implement a program that allows you to record and analyze rainfall indices in **three cities** (Braga, Porto, Lisbon) over **6 months** (from January to June).

1.1 Create a two-dimensional list called **rainfall**, where:

- Each **line** represents a **city**;
- Each **column** represents a **month**.

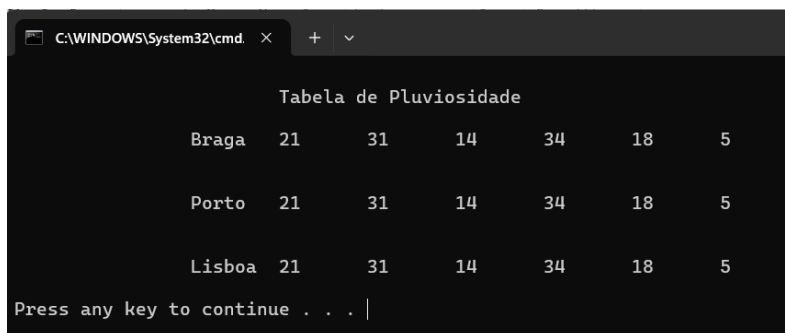
1.2 Your application should:

- Ask the user to indicate the rainfall values (in millimeters of rain).
- If you prefer, you can generate the monthly rainfall values for each city, randomly, between 0 and 100;
- Call a function **showData()** that shows the data you entered, in table format;

```
Pluviosidade em Braga
Janeiro :34
Fevereiro:39
Março   :27
Abril   :44
Maio    :21
Junho   :12
```

```
Pluviosidade em Porto
Janeiro :35
Fevereiro:45
Março   :39
Abril   :54
Maio    :19
Junho   :12
```

```
Pluviosidade em Lisboa
Janeiro :|
```



The screenshot shows a Windows command prompt window with the title 'C:\WINDOWS\System32\cmd. x'. Inside the window, a table titled 'Tabela de Pluviosidade' is displayed. The table has three rows for cities (Braga, Porto, Lisboa) and six columns for months (January to June). The data values are: Braga (21, 31, 14, 34, 18, 5), Porto (21, 31, 14, 34, 18, 5), and Lisboa (21, 31, 14, 34, 18, 5). Below the table, the prompt 'Press any key to continue . . . |' is visible.

	Jan	Fev	Mar	Abr	Maio	Jun
Braga	21	31	14	34	18	5
Porto	21	31	14	34	18	5
Lisboa	21	31	14	34	18	5

- Call a function named **dataAnalysis()** that calculates and *returns*:
 - The **average rainfall of** each city over the 6 months;
 - The **average rainfall** for each month (considering all cities);

3. Develop a program that allows you to manage a small parking car with 3 rows, each one with 5 parking places. The park layout is similar to that shown in the image:

1	2	3	4	5	Fila 1
1	2	3	4	5	Fila 2
1	2	3	4	5	Fila 3

When the program starts, all parking places must be free. Your program should start through a menu with the following options:

MENU

- 1 – Car Entrance
- 2 – Car Exit
- 3 - Park State
- 0 - Exit

- **Car Entrance:** must occupy the first parking place that is free, starting from row 1 (place 1) and ending in row 3 (place 5). You must indicate on the console the position of the seat to be occupied.
If all parking seats are occupied, please show the message "Park Complete!".
- **Car Exit:** The user must indicate the position of their car in the car park: row and place. The indicated position must be occupied, changing its status to free.
- **Park Status:** this option should show the park's occupancy. It should show the number of occupied places and the number of free places in the park.